

Project Initialization and Planning Phase

Date	26 june 2026
Team ID	SWTID-2026-8613
Project Name	Personalized Networking Assistant
Maximum Marks	3 Marks

Define Problem Statements (Customer Problem Statement Template)

Create a problem statement to understand your customer's point of view. The Customer Problem Statement template helps you focus on what matters to create experiences people will love. A well-articulated customer problem statement allows you and your team to find the ideal solution for your customers' challenges. Throughout the process, you'll also be able to empathize with your customers, which helps you better understand how they perceive your product or service.

Problem Statement

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	Networking event attendee	Start meaningful professional conversations	Finding relevant conversation topics with strangers is difficult	Event themes vary and people often lack confidence or background knowledge	Confident and prepared with AI-generated, verified conversation starters

Model Research & Selection (Integrated)

Introduction

Selecting the right AI model is critical because application quality depends on response accuracy, speed, and user experience. Two NLP capabilities were required: event theme understanding and conversation generation.

Theme Extraction – DistilBERT

DistilBERT was selected because it supports zero-shot classification, allowing classification into user-defined categories without training a custom dataset. It provides an excellent balance of accuracy and speed, requires less memory and computation than BERT, and is suitable for deployment on standard hardware.

Conversation Generation – GPT-2 Small

GPT-2 Small was chosen because it generates short, natural, context-aware conversation starters while remaining lightweight. It requires fewer computational resources, runs on CPU hardware, integrates easily with the Hugging Face Transformers library, and is simple to deploy.

Why These Models Were Selected

Together, DistilBERT and GPT-2 Small provide the best balance of performance, speed, scalability, deployment efficiency, and computational cost.

Conclusion

DistilBERT accurately extracts event themes, while GPT-2 Small generates meaningful conversation starters. Together they satisfy the functional requirements of the Personalized Networking Assistant by providing fast, reliable, and practical AI assistance.